

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the structure of two lipid molecules, **X** and **Y**.

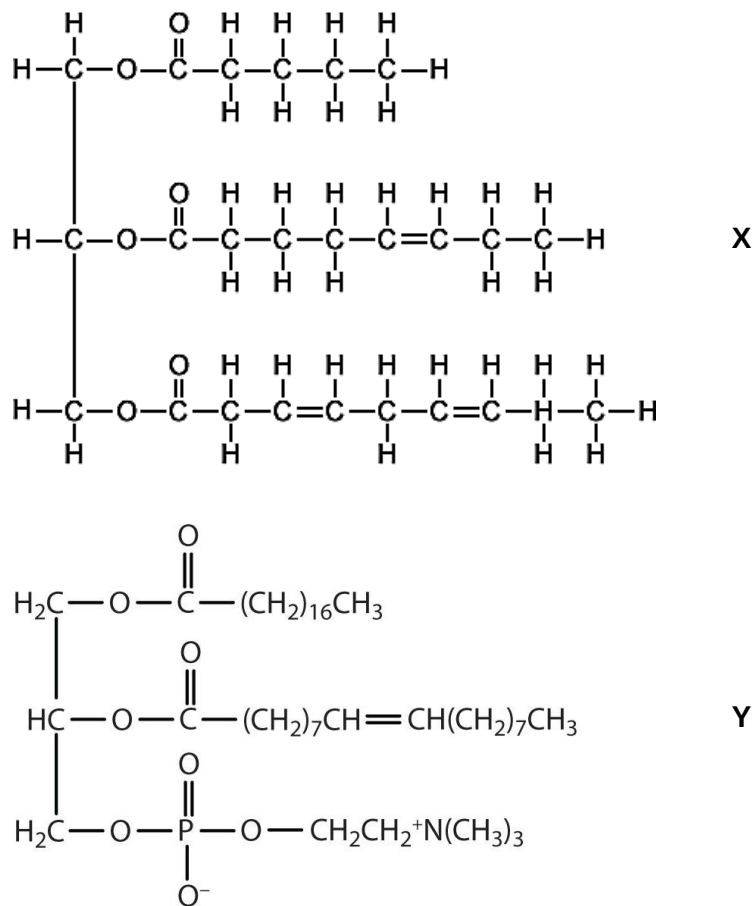


Fig. 1.1

- (a) With reference to Fig. 1.1,
- (i) describe **two** structural features which are similar between molecules **X** and **Y**.

[2]

- (ii) explain how the structure of molecule **Y** is related to bilayer arrangement in cell membrane.

[3]

Other than lipids, proteins are also the building blocks of life. Every cell in the human body contains protein. The basic structure of protein is a chain of amino acids.

- (b) Describe how a dipeptide is formed from two amino acids.

[2]

One of the important protein molecules in the human body is collagen. Collagen is the main structural protein. It strengthens the tendons and supports the skin and internal organs. Bones and teeth are made by adding mineral crystals to collagen. Collagen is made up of many tropocollagen subunits. Each tropocollagen subunit is composed of three polypeptide chains, wound together in a tight triple helix. Fig. 1.2 shows only a segment of the collagen.

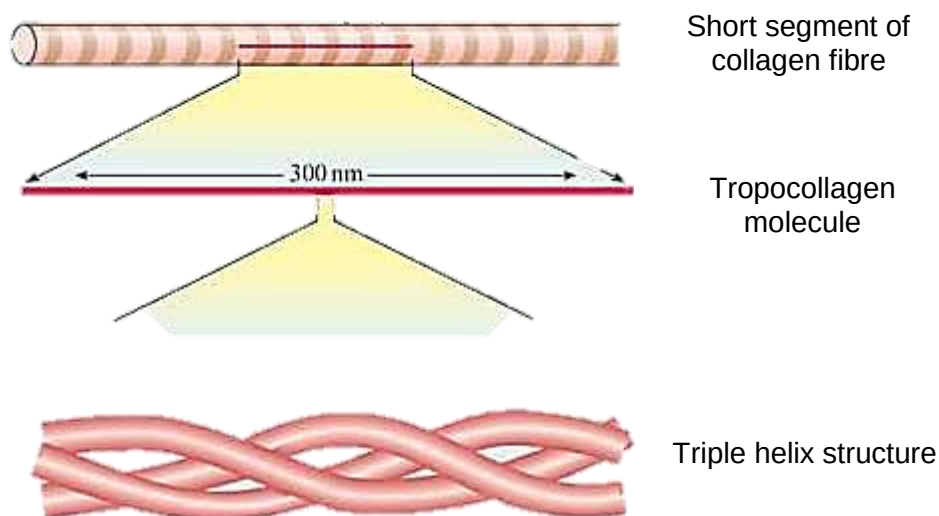


Fig. 1.2

(c) Explain why collagen is described as a fibrous protein.

[2]

Cellulose is an important structural component of the primary cell wall of green plants while collagen is the main structural protein in the extracellular space in the various connective tissues in animal bodies.

(d) Describe **two** structural differences between collagen and cellulose.

[2]

[Total: 11]

- 2 DNA is found mostly in the cell nucleus, but another type of nucleic acid, RNA, is common in the cytoplasm. Watson and Crick proposed that RNA must copy the *template* found in the nucleus and carry it out to the cytoplasm, where proteins are synthesized. Crick also predicted the existence of an "adaptor" molecule that reads the genetic code and selects the appropriate amino acids to add to a growing polypeptide chain. This proposed flow of genetic information is known as the "Central Dogma".

As it turned out, several types of RNA are involved in the utilization of genetic information. In the nucleus, the code is "transcribed," or copied, into a messenger RNA (mRNA) molecule. In the cytoplasm, the mRNA code is "translated" into amino acids. Translation is orchestrated at the ribosome, itself partly composed of RNA, with transfer RNA playing the role of adaptor.

RNA is an intermediary between DNA and protein. (n.d.). Retrieved July 03, 2016, from <http://www.dnafb.org/21>

- (a) Describe the functions of the nuclear envelope that encloses the nucleus.

[2]

- (b) With reference to the information given above,

- (i) suggest how the structure of the adaptor molecule is suitable for its role.

[2]

- (ii) explain what is meant by the term *template*.

[3]

- (c) Explain why the unwinding of the DNA double helix promotes transcription.

[1]

[Total: 8]

- 3 Fig. 3.1 shows the course of a human immunodeficiency virus (HIV) infection. The graph shows HIV RNA copies and CD4⁺ T lymphocytes counts over course of the infection.

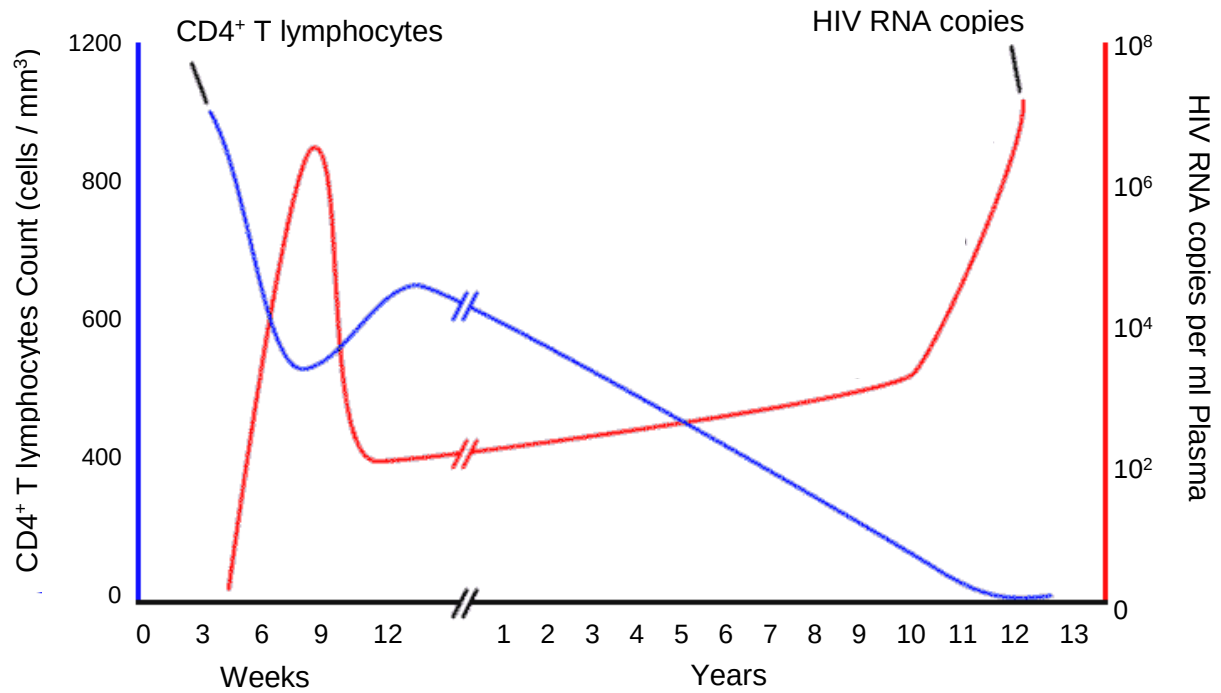


Fig. 3.1

- (a) (i) Outline how HIV infects the cell.

[4]

- (ii) With reference to Fig. 3.1, explain the course of HIV infection from week 1 to week 12.

[2]

- (b) Fig. 3.2 shows how a defective lambda phage could be formed.

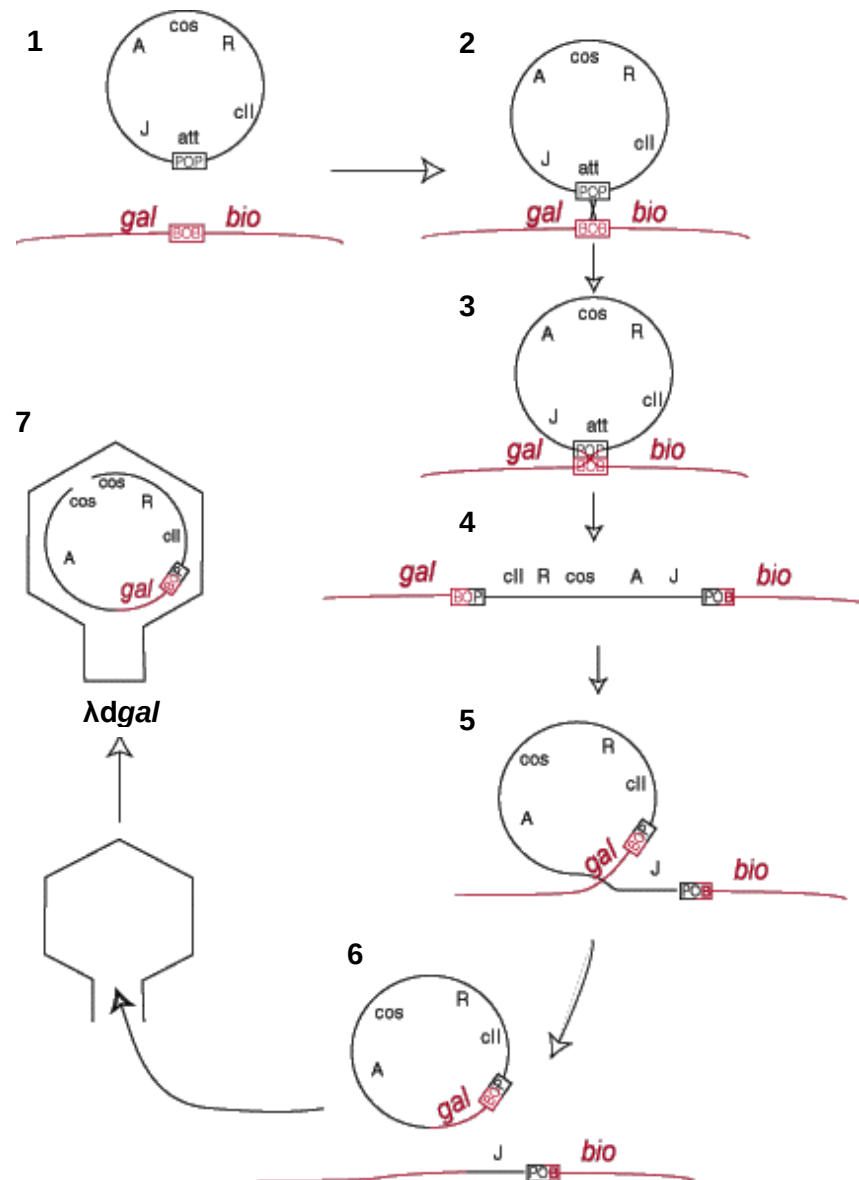


Fig. 3.2

- (i) What is a lambda (λ) phage?

[1]

- (ii) Name **two** important requirements necessary for the DNA of lambda phage to be integrated into the host's DNA as shown in stages 2 to 4.

[2]

- (iii) With reference to Fig. 3.2, suggest and explain a possible outcome of the host DNA from this infection when the defective phage shown in stage 7 infects another host cell with *gal*⁻ allele.

[2]

[Total: 11]

- 4 The TATA box binding protein (TBP) attaches to a specific DNA sequence found in the regulatory region upstream of the insulin gene. TBP, along with a variety of TBP-associated factors, form the transcription initiation complex.

Fig. 4.1 shows TBP attached to DNA.

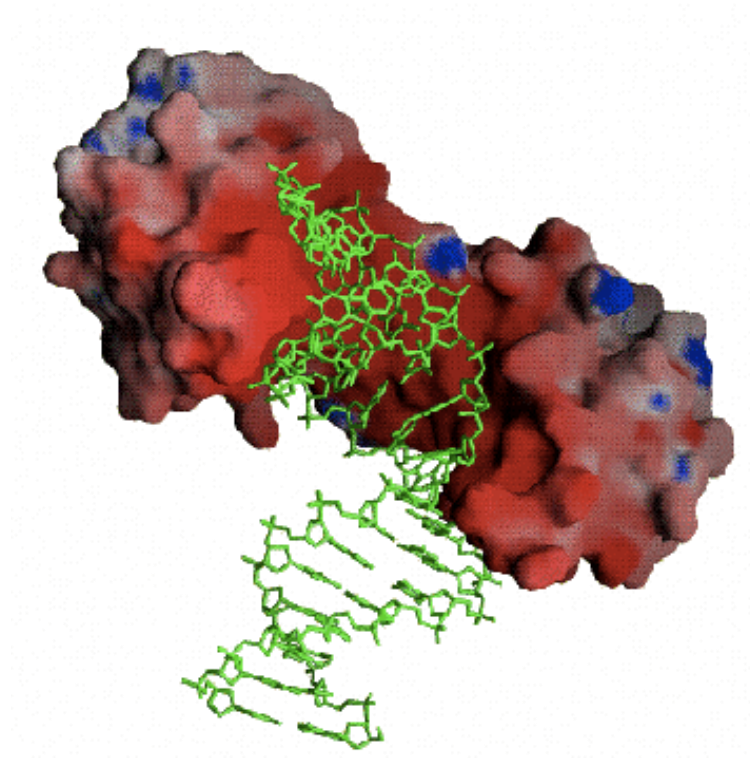


Fig. 4.1

TBP. (n.d.). Retrieved July 03, 2016, from <https://ghr.nlm.nih.gov/gene/TBP>

(a) With reference to the information given and Fig. 4.1,

(i) state the name of the *regulatory region of DNA*.

[1]

(ii) suggest how TBP will bind to this *regulatory region*.

[1]

(iii) suggest how TBP will initiate transcription.

[1]

Fig. 4.2 shows how a mature insulin mRNA is synthesised via post-transcriptional modification processes.

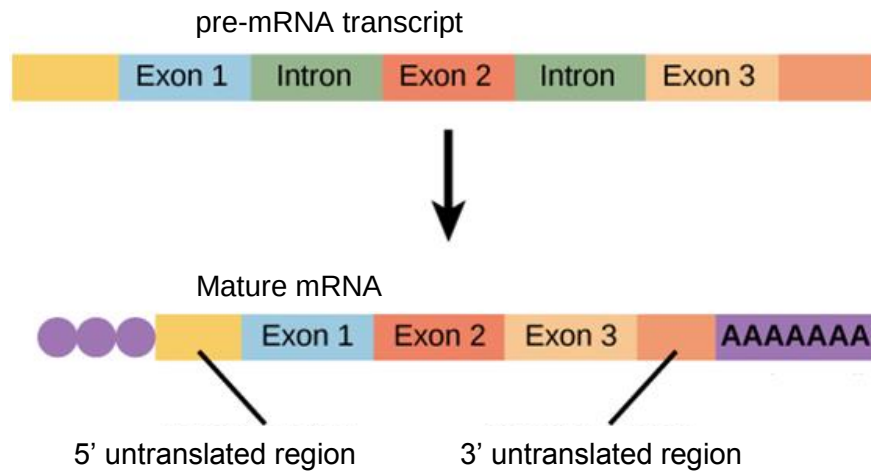


Fig. 4.2

- (b)** Explain why these post-transcriptional modifications are necessary.

[3]

- (c)** The mature insulin is translated to form a preproinsulin which is an inactive precursor.

Describe one specific post-translational modification that converts it to its active form.

[2]

[Total: 8]

- 5** A transcription factor is a protein that promotes transcription at a particular locus. The gene *Ets 2* codes for a transcription factor, and is located on human chromosome 21. The gene is expressed in cartilage and bone tissues during development. In mice, *Ets 2* is located on chromosome 16 and mice with trisomy 16 show similar skeletal abnormalities to humans with Down's syndrome.

(a) (i) Explain the genetic basis of Down's syndrome in humans.

[2]

(ii) Explain how a trisomy, such as trisomy 16 in mice, can occur in a zygote.

[3]

In order to investigate the possible role of *Ets 2* in Down's syndrome, transgenic mice have been produced which carry an extra copy of mouse *Ets 2*. These mice show similar skeletal abnormalities to mice with trisomy 16. Tissues from the transgenic mice produced up to 1.8 times more *Ets 2* messenger RNA than control mice.

(iii) Suggest why these transgenic mice and mice with trisomy 16 show similar abnormalities.

[2]

Besides chromosomal mutations, gene mutations also occur in humans. Haemophilia A is characterized by deficiency in coagulation factor VIII, which results in prolonged oozing after injuries, tooth extractions, or surgery and renewed bleeding after initial bleeding has stopped. *F8* gene codes for the coagulation factor VIII.

In severe haemophilia A, spontaneous joint bleeding is the most frequent symptom. Serious complications can result from bleeding into the joints, muscles, brain, or other internal organs. Severity of symptoms can vary.

- (b)** Explain how substitution of bases in the *F8* gene could cause different levels of severity of symptoms in Haemophilia A.

[3]

[Total: 10]

- 6** In summer squash, there are two pairs of alleles that determine fruit colour. The two genes are known to assort and segregate independently.

Two white-fruited plants are crossed. Both parents are known to be heterozygous for both genes. The cross produces the following offspring: 20 green-fruited plants, 58 yellow-fruited plants, and 218 white-fruited plants.

- (a)** State the name for this type of interaction between the two genes.

_____ [1]

- (b)** Using the symbols D, d and E, e to draw a genetic diagram to explain the results given in the above information.

[4]

- (c) Using enzyme production as a biochemical basis for the fruit colour in squash, explain how the two genes interact to produce the various phenotypes.

[3]

A chi-squared test was carried out on the results of the cross.

Phenotype	Observed (O)	Expected (E)	$\frac{(O - E)^2}{E}$
green-fruited plants	20	19	0.1216
yellow-fruited plants	58	56	0.1126
white-fruited plants	218	222	0.0721
			$\chi^2 = 0.306$

Part of the critical values of the chi-squared distribution is shown below.

Degree of freedom	Probability								
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345

- (d) Explain how the chi-squared calculated value of 0.306 supports the hypothesis that the two genes assorted and segregated independently.

[3]

[Total: 11]

- 7 It is important that the body's internal environment is controlled to maintain relative constancy of the internal environment despite external environmental changes. Maintaining a constant internal environment is called homeostasis. The nervous and endocrine systems both work to bring about this but their response patterns are different. Both the systems use chemical messengers to signal cells, but the speed at which these messages are transmitted is different.

(a) Outline how the body receives information from the external environment to bring about a response. [3]

[3]

Fig. 7.1 is a schematic diagram of the interaction between the different systems with the external environment.

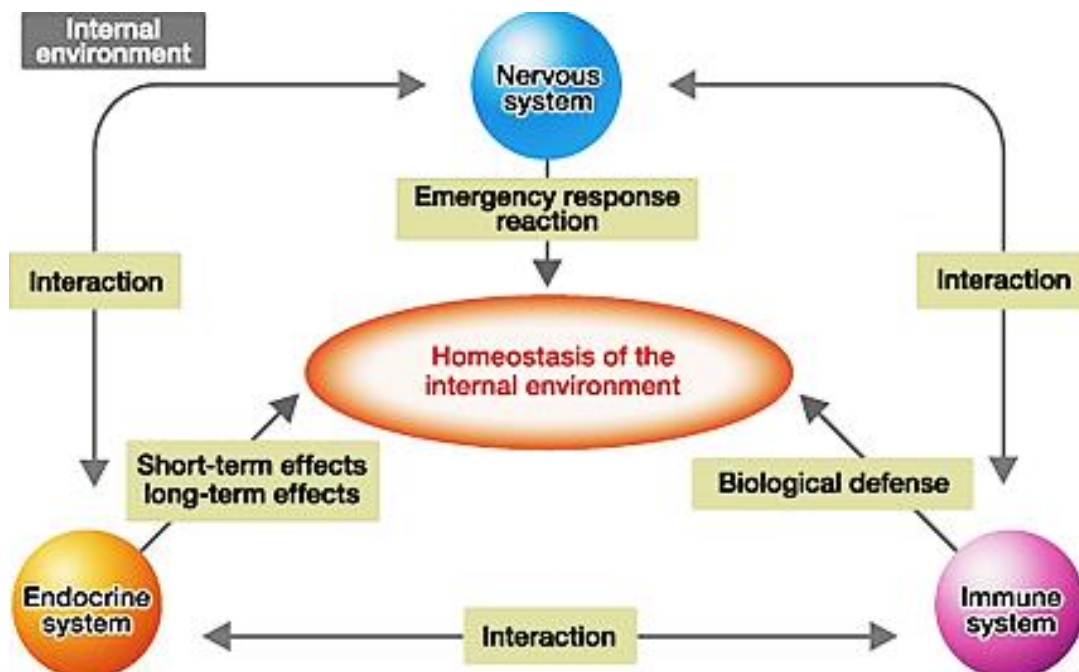


Fig. 7.1

- (b) With reference to Fig. 7.1, describe the important differences between endocrine and nervous systems.

[2]

Fig. 7.2 shows the steps needed to transmit a signal across a junction known as a synapse between two neurones.

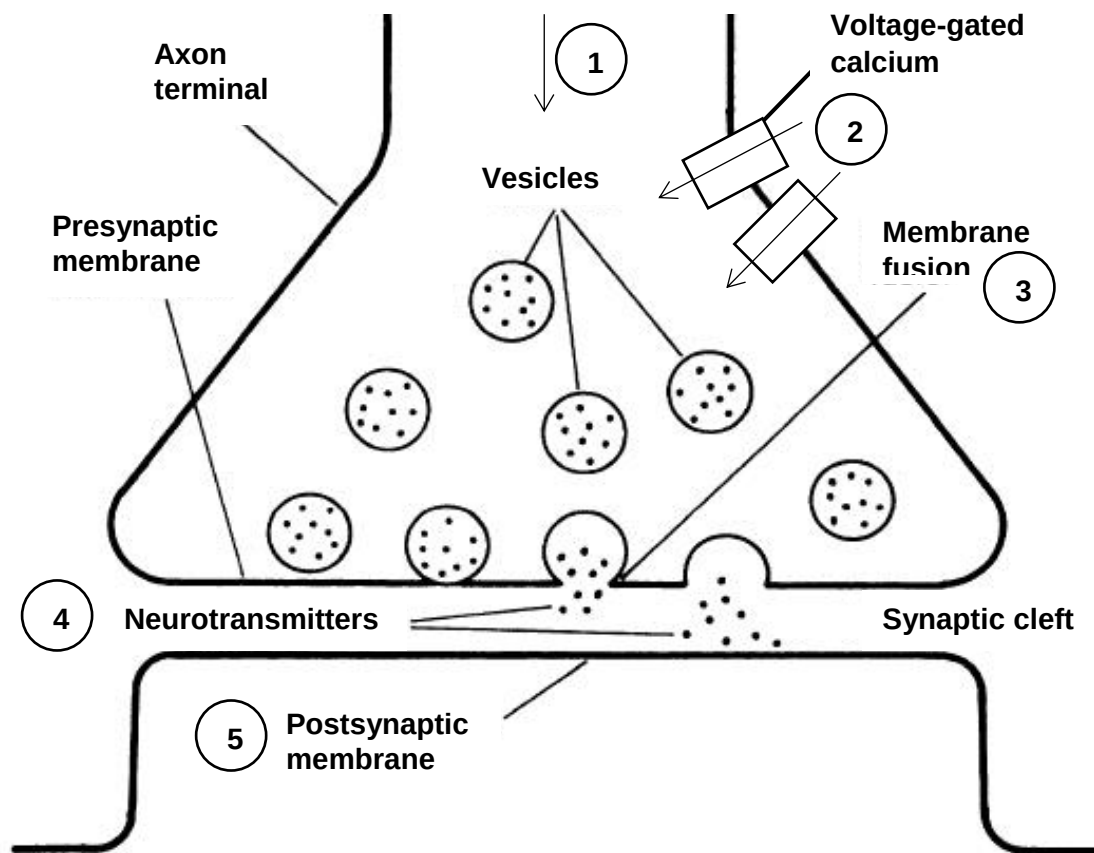


Fig. 7.2

- (c) (i) Explain which steps shown in Fig. 7.2 will result in synaptic delay.

[2]

- (ii) Describe the roles of calcium ions in signal transmission across the synapse.

[2]

The synaptic knob releases a chemical transmitter, acetylcholine. Nerve gas prevents the breakdown of this chemical.

- (d) Suggest and explain one effect of nerve gas on the transmission of nerve impulse across the synapse.

[1]

[Total: 10]

- 8 Anoles are small, colour-changing lizards that are abundant in the Caribbean Islands. Hundreds of species live on the six islands of the Caribbean. Biologist Jonathan Losos has discovered the traits that enable dozens of anole species to adapt to different vertical niches in the forest of the islands.

While differences in limb length, body shape, and toepad size allow different species to flourish on the ground, on thin branches, or high in the canopy, changes in other characters, such as their colourful dewlaps, have played a key role in reproductive isolation and the formation of new species.

Fig. 8.1 shows a picture of *Anolis grahami*, commonly called Graham's anole. This is a species originally native to the island of Jamaica. Jamaica, part of the Caribbean islands, has a lush topography of mountains, rainforests and reef-lined beaches. Graham's anole is one of eight anole species found on Jamaica.



Fig. 8.1

- (a) Complete Table 8.1 to show the classification of *Anolis grahami*.

Kingdom	
	Chordata
	Reptilia
	Squamata
Family	Polychrotidae
Genus	
Species	<i>Anolis grahami</i>

[3]

Differences in the DNA sequences that encode cytochrome *b* in the eight species of Jamaican anoles have been measured.

- (b) Suggest why the cytochrome *b* gene is used to measure changes in DNA sequences in closely related species.

[2]

Fig. 8.2 shows differences in cytochrome *b* DNA plotted against time since colonisation (My).

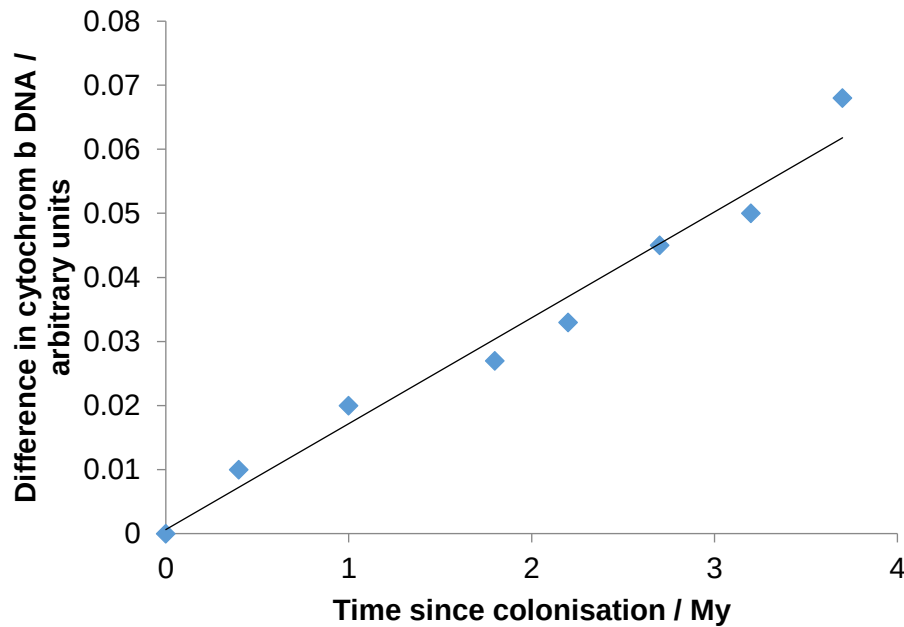


Fig. 8.2

- (c) Describe how these changes in DNA sequence of cytochrome *b* supports the neutral theory of evolution.

[3]

Using similarities in the banding of their chromosomes, it is possible to construct a phylogenetic tree for the evolution of these *Anolis* lizards.

Fig. 8.3 shows the phylogenetic tree for eight species of the *Anolis* lizards.

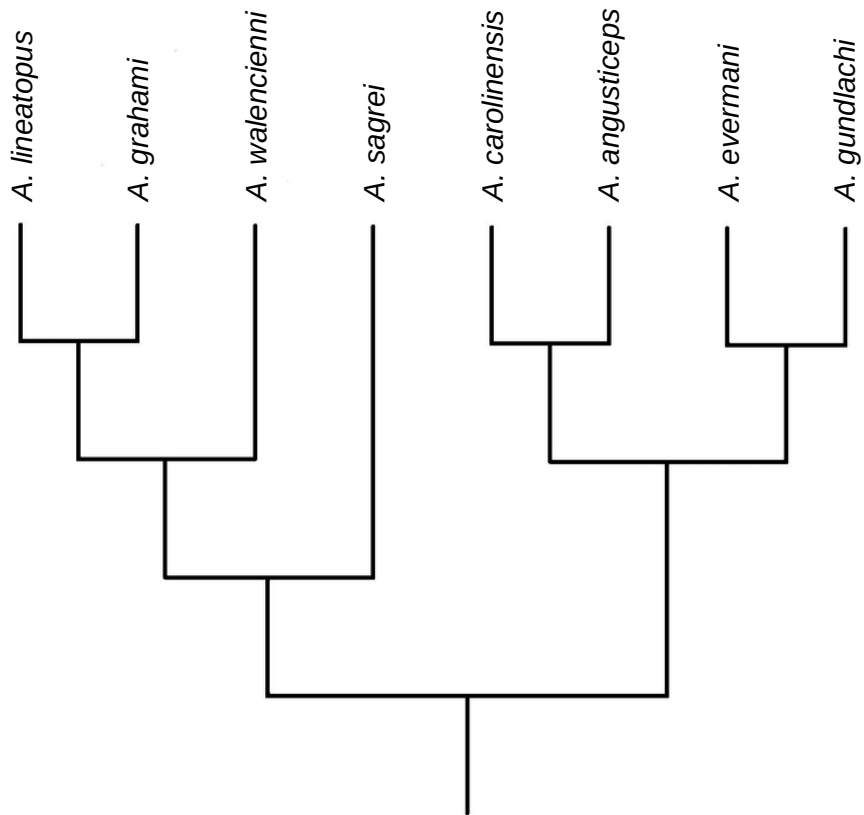


Fig. 8.3

(d) Explain the relationship between classification and phylogeny.

[3]

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9** **(a)** Outline the three phases of the Calvin cycle. [6]
- (b)** Explain the need for the production of genetically identical cells and fine control of replication. [8]
- (c)** Describe the roles of NAD⁺ in aerobic respiration. [6]

[Total: 20]

- 10** **(a)** Outline the process of binary fission in bacteria. [6]
- (b)** Explain the effects of competitive and non-competitive inhibitors on the rate of enzyme activity. [6]
- (c)** Describe the causes of genetic variation in a population. [8]

[Total: 20]

Suggested Answers with marking feedback

- 1 Fig. 1.1 shows the structure of two lipid molecules, X and Y.

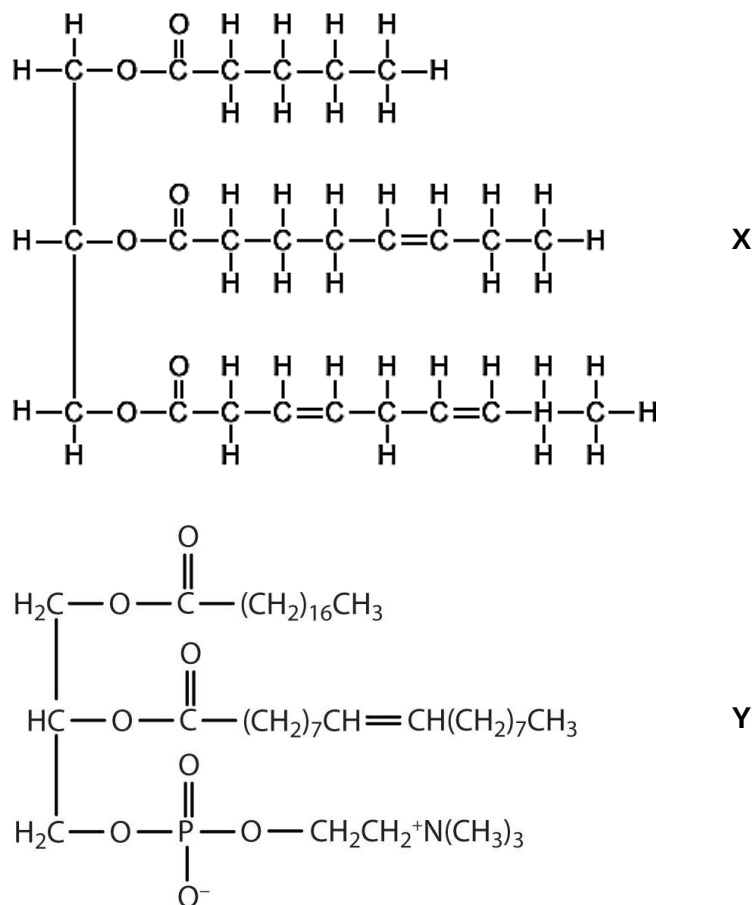


Fig. 1.1

- (a) With reference to Fig. 1.1,

- (i) describe **two** structural features which are similar between molecules **X** and **Y**.
1. Both molecule X (triglyceride) and molecule (phospholipid) contains a **glycerol molecule**.
 2. Both molecules contain **at least 1 ester bond**. (Reject if student write three ester bonds as phospholipid only has 2 ester bond.)
 3. Both molecules contain **at least 1 chain of hydrocarbon**. (Reject if student write three hydrocarbon chains as phospholipid only has 2 ester bond.)
 4. Both molecules contain **1 chain of unsaturated hydrocarbon**.
 5. Both molecules contain **carbon, oxygen and hydrogen atoms**.

Any **2**

- (ii) explain how the structure of molecule **Y** is related to bilayer arrangement in cell membrane.
1. The phosphate head is polar / hydrophilic thus able to form hydrogen bonds with water molecules in the aqueous environment.
 2. The fatty acid chains are long and non-polar / hydrophobic thus resulting them facing away from the aqueous environment to form the hydrophobic core.
 3. Unsaturated fatty acid chains contains carbon-carbon double bonds (C=C), resulting in the formation of kinks that prevents the close packing of phospholipids.

Other than lipids, proteins are also the building blocks of life. Every cell in the human body contains protein. The basic structure of protein is a chain of amino acids.

(b) Describe how a dipeptide is formed from two amino acids.

1. The carboxyl group (-COOH) of one amino acid is linked to the amino group (-NH₂) of another amino acid through condensation / removal of a molecule of water.
2. A peptide bond (accept amide linkage) (-CONH-) is formed between the two amino acids.

One of the important protein molecules in the human body is collagen. Collagen is the main structural protein. It strengthens the tendons and supports the skin and internal organs. Bones and teeth are made by adding mineral crystals to collagen. Collagen is made up of many tropocollagen subunits. Each tropocollagen subunit is composed of three polypeptide chains, wound together in a tight triple helix. Fig. 1.2 shows only a segment of the collagen.

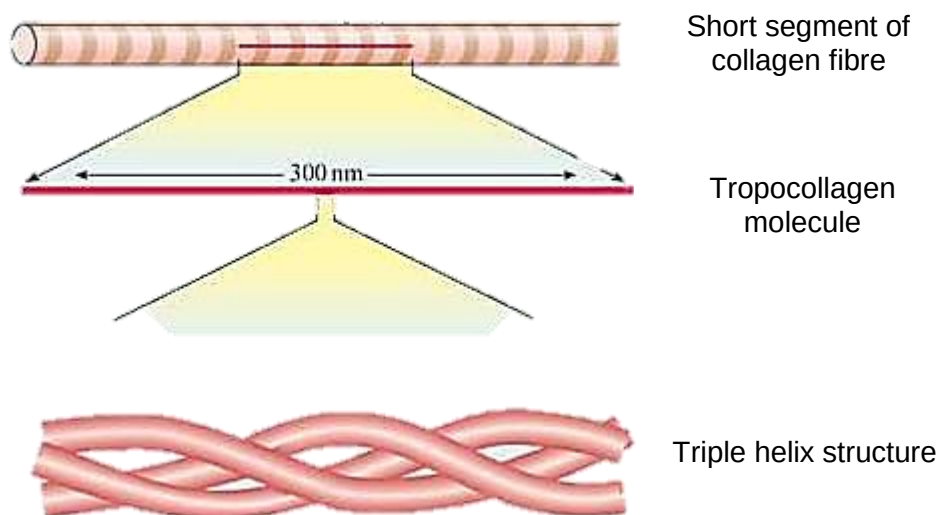


Fig. 1.2

(d) Explain why collagen is described as a fibrous protein.

1. Each polypeptide chain comprises of a primary structure of regular repetitive sequences of proline-hydroxyproline-glycine.
2. Each polypeptide chain has a predominantly secondary helical structure.
3. Three polypeptide chains wound around each other to form a triple helix / tropocollagen.
4. This allows the tropocollagen to form long fibres which may run parallel to one another, linked by cross-bridges.
5. Tropocollagen is insoluble in water because there are no free polar groups to form hydrogen bonds with water molecules OR all the $-NH$ groups on glycine residue and all the $-CO$ groups of proline residues are involved in hydrogen bonding to stabilise the triple helix OR proline and glycine have non-polar R groups that are unable to interact with water molecules.

Cellulose is an important structural component of the primary cell wall of green plants while collagen is the main structural protein in the extracellular space in the various connective tissues in animal bodies.

(d) Describe **two** structural differences between collagen and cellulose.

Cellulose	Collagen
Cellulose is a polymeric unit made up of <u>glucose</u> .	Collagen is a polymeric unit made up of <u>amino acids</u> .
Compose of <u>one repeat unit</u> of glucose. (only β -glucose)	Compose of <u>three repeat units</u> of amino acids (Gly-Pro-X).
Monomers of cellulose (glucose) are linked via <u>glycosidic bonds</u> .	Monomers of collagen (amino acids) are linked via <u>peptide bonds</u> .
Cellulose fiber is made up of <u>a single straight and unbranched chain</u> of glucose polymer.	Collagen fiber is composed of <u>three helical polypeptide chain</u> twisted together to form a triple helix.
H-bonding maintains the bundles of fiber of cellulose. H bonding is between <u>OH groups</u> of the beta glucose.	H-bonding maintains the bundles of fiber of collagen. H bonding is between <u>C=O and H-N</u> groups of the amino acids.
Comparing it with the bundle of fibres in cellulose, there are <u>no covalent cross-links</u> between the fibres.	Within the tropocollagen, some <u>covalent bonds</u> are present to maintain the triple helix structure.

Accept any other valid answers such as rotation of beta-glucose and no rotation of amino acids.

Any two rows; students' answer must be point-to-point comparison to be awarded credit.

[Total: 11]

- 2 DNA is found mostly in the cell nucleus, but another type of nucleic acid, RNA, is common in the cytoplasm. Watson and Crick proposed that RNA must copy the *template* found in the nucleus and carry it out to the cytoplasm, where proteins are synthesized. Crick also predicted the existence of an "adaptor" molecule that reads the genetic code and selects the appropriate amino acids to add to a growing polypeptide chain. This proposed flow of genetic information is known as the "Central Dogma."

As it turned out, several types of RNA are involved in the utilization of genetic information. In the nucleus, the code is "transcribed," or copied, into a messenger RNA (mRNA) molecule. In the cytoplasm, the mRNA code is "translated" into amino acids. Translation is orchestrated at the ribosome, itself partly composed of RNA, with transfer RNA playing the role of adaptor.

RNA is an intermediary between DNA and protein. (n.d.). Retrieved July 03, 2016, from <http://www.dnafb.org/21>

- (a) Describe the functionss of the nuclear envelope that encloses the nucleus. [2]

- (1) To separate nucleoplasm from cytoplasm/ compartmentalisation/ served as a barrier to provide a localised and optimum environment in the nucleus for processes such as transcription that occur in the nucleus

A: creates an enclosed environment; contained within the nucleus to prevent DNases/endonucleases/exonucleases from breaking down the DNA.

R: suitable environment for cellular activity/ compartmentalisation of the nucleolus/ work more efficiently without stating optimum/localised environment; the simple stating down of preventing DNA from being degraded/digested without stating that it is the action of relevant enzymes and etc

- (2) To regulate/control movement of materials between nucleus and cytoplasm into the cytoplasm or ribosomal subunits and nucleotides into the nucleoplasm, e.g. movement of mRNA

A: have pores to allow the movement of / allows only certain molecules to pass through such as mRNA or small molecules into the cytoplasm for translation (taught in DNA and Genomics)

R: allows only small molecules (concept is certain molecules or specific molecules) to pass through; allows only certain molecules to pass through without the mention of nuclear pores or selectively permeable.

R: separates the process of transcription and translation; there must be the concept that this only occurs in eukaryotic cells. So if students did not state clearly that this occurs in eukaryotic cells only, no marks will be given

- (b) With reference to the information given above,

- (i) suggest how the structure of the adaptor molecule is suitable for its role.

3' CCA end/stem that the specific amino acid will be binded to the other end of the tRNA is made up of three bases known as an anticodon that is complementary/binds to the codons of the mRNA

R: do not accept CCA only without the 3' 3' OH end

(ii) explain what is meant by the term *template*.

(1) In the synthesis of RNA molecules, the template used is DNA/ antisense/ non-coding DNA.

R: sense/coding strand of DNA (this strand is not used)

(2) The DNA base sequence is used to form complementary base pair using free ribonucleotides/ encode for the ribonucleotide sequence.

(3) The template serves to ensure that the same genetic information is passed to amino acid sequence every time a gene is transcribed and translated.

Or:

This ensure that the genetic code is passed down to mRNA/sequence of mRNA through transcription and to the amino acid sequence (reject protein sequence) through translation.

(c) Explain why the unwinding of the DNA double helix promotes transcription.

RNA polymerase/ can bind to the promoter region. Or the transcription factors can bind to the TATA box.

[Total: 8]

- 3 Fig. 3.1 shows the course of an HIV infection. The graph shows HIV copies and CD4 counts over course of a typical HIV infection

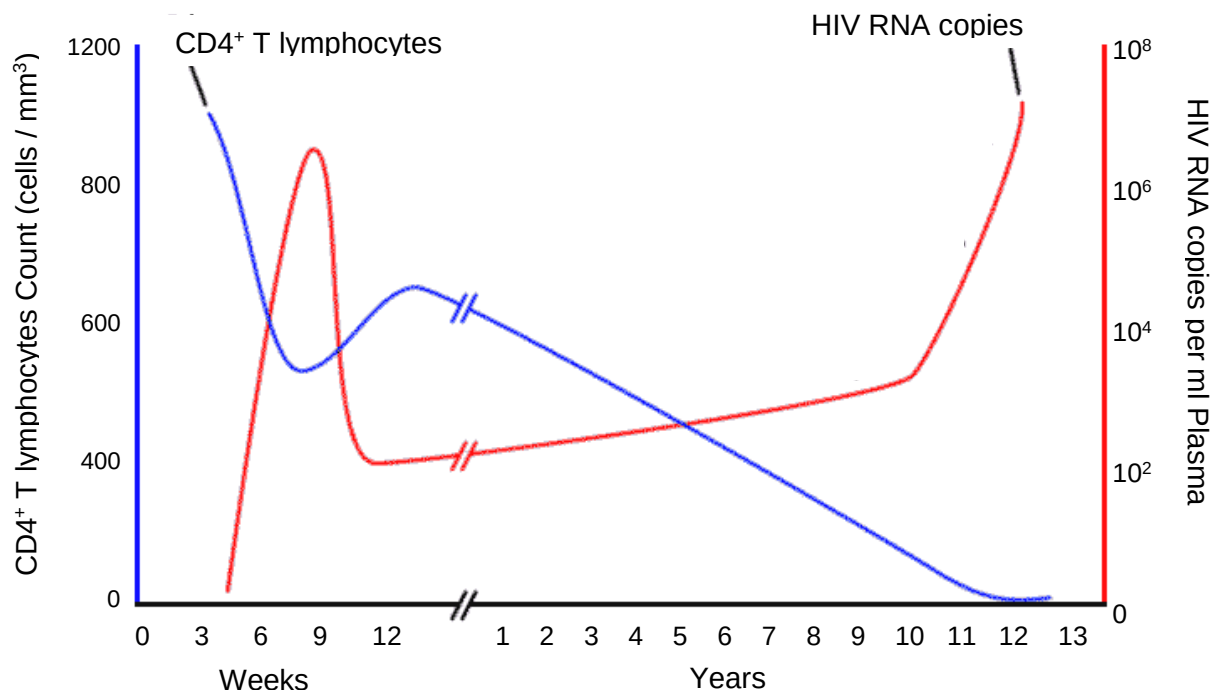


Fig. 3.1

- (a) (i) Outline how the human immunodeficiency virus infects the cell.
1. Complementary binding of the virus spikes /gp 120 to specific host receptors on the CD4⁺ T cells and HIV envelope fuses with the host cell membrane envelope
 2. RNA released into cytoplasm where reverse transcriptase uses viral + sense RNA as template to synthesise a strand of cDNA and then form a double stranded viral DNA.
 3. The DNA enters the nucleus and integrase catalyses the integration into the chromosome DNA to form a provirus
 4. The provirus DNA is transcribed to form viral mRNA and also viral mRNA can be used as template for translation to viral proteins
 5. The HIV protease cleaves the long chains of polyproteins and assembly of new HIV with the viral genome and proteins before budding out of host cells.

- (ii) With reference to Fig. 3.1, explain the course of an HIV infection from week 1 to week 12.
- 1 From 3 to 8 weeks, HIV RNA copies increase to 10^7 copies per ml while CD4 T cells dropped to 550 cells/mm³. This is because the virus replicating rapidly and the number of T cells in the blood decreases as the infected cells destroyed after a couple of days.
 - 2 From 8 weeks to 12 weeks, HIV RNA copies decrease to 10^2 copies per ml while CD4 T cells increase to 620 cells/mm³. This is because the body's natural response to an infection is to fight infected cells, Virus are killed by the immune system/antibodies and the CD4 cells lost are replaced.

(b) Fig. 3.2 shows how a defective lambda phage could be formed.

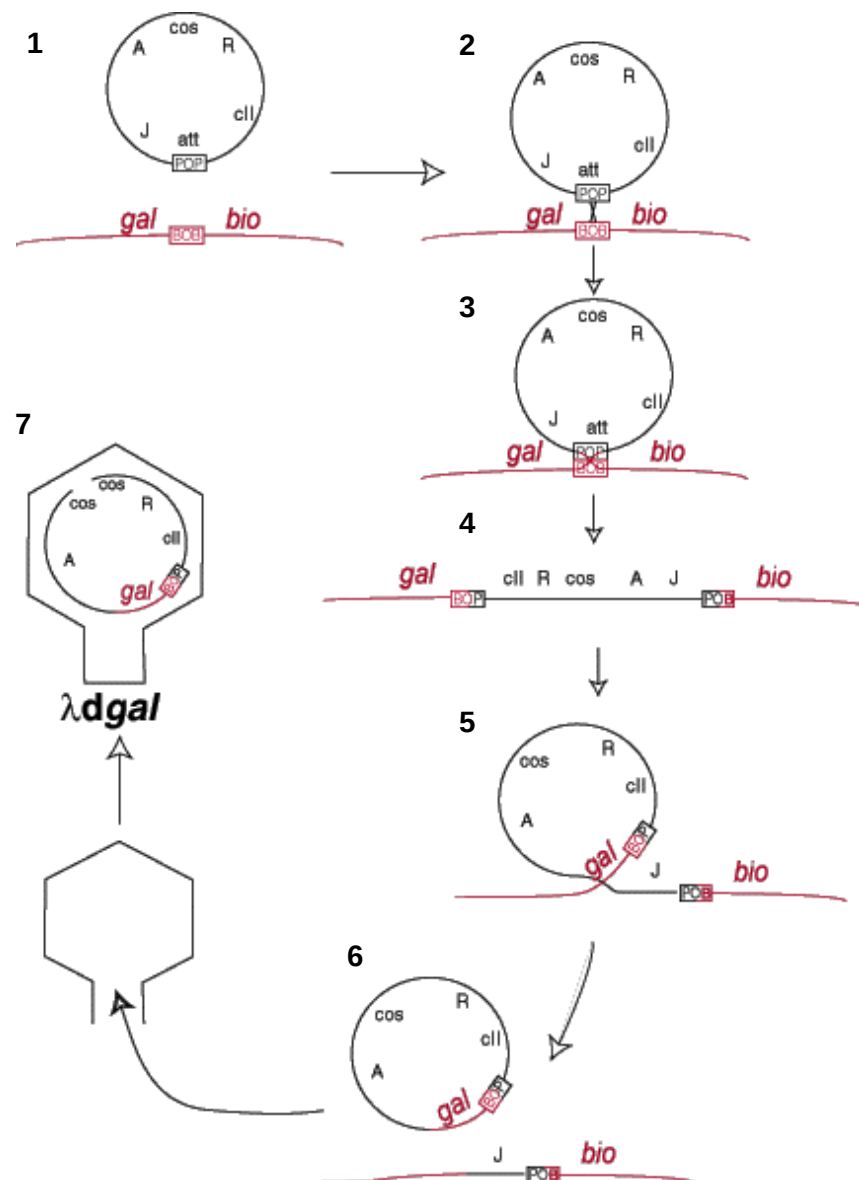


Fig. 3.2

(i) What is a lambda (λ) phage?

It is a temperate phage as it is able to go through a **lytic cycle** and destroy host cell or enter a **lysogenic stage** where the viral integrates into the bacteria's DNA as a prophage. [1]

(ii) Name **two** important requirements necessary for the DNA of lambda phage to be integrated into the host's DNA as shown in stages 2 to 4.

- 1 repression of all lytic functions by lambda repressor
- 2 requires enzymes , lambda integrase to allow integration
- 3 a special site /site-specific recombination) in the phage DNA and host DNA. The site is called *att* (*attachment* site)

[any 2]

(iii) With reference to Fig. 3.2, suggest and explain a possible outcome of the host DNA from this infection when the defective phage shown in stage 7 infects another host cell with *gal*⁻ allele.

- 1 The *gal* allele from the defective phage can be incorporated into the host chromosome by **homologous recombination** due to crossover between the phage allele and chromosome allele.
- 2 The crossover involves **Host's *gal*⁻ allele in exchange for the phage *gal*⁺ allele** → **result in** recipient converted to *gal*⁺ by specialized transduction.

[Total: 11]

- 4 The TATA box binding protein (TBP) attaches to a specific DNA sequence found in the regulatory region upstream of the insulin gene. TBP, along with a variety of TBP-associated factors, form the transcription initiation complex.

Fig. 4.1 shows TBP attached to DNA.

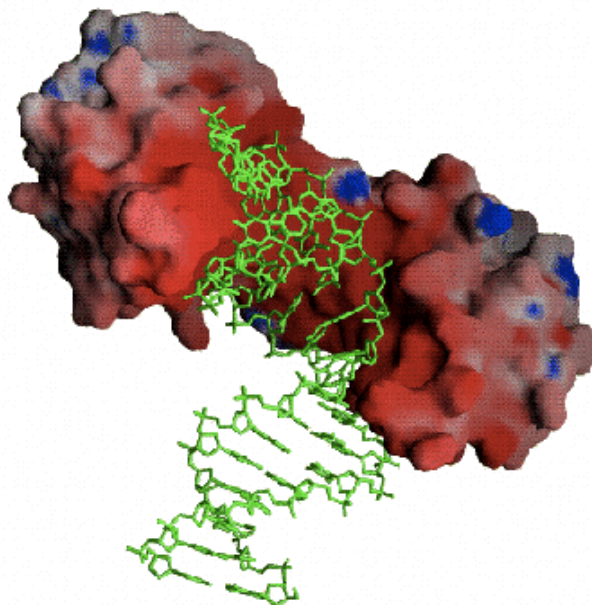


Fig. 4.1

TBP. (n.d.). Retrieved July 03, 2016, from <https://ghr.nlm.nih.gov/gene/TBP>

- (a) With reference to the information given and Fig. 4.1,

- (i) state the name of the *regulatory region*.

Promoter; reject TATA box (read the question, TATA box can only be PART of the regulatory region indicating that the regulatory region must be the promoter region)

- (iii) suggest how TBP will bind to this *regulatory region*.

The **binding site** of the TBP is **complementary to the specific base sequence/ sequence**, being TATA box of the promoter region. [1]

R: active site of TBP; it is not an enzyme.

R: complementary to DNA or DNA structure, it must be complimentary to the regulatory sequence only, or else the proteins can bind to all parts of the DNA instead.

- (iv) suggest how TBP initiates transcription.

When TBP binds to a TATA box, it distorts the DNA/ place strain on the two DNA strands and partially unwinds the DNA double helix.
(best answer because this can be seen from the diagram)

Fig. 4.2 shows how a mature insulin mRNA is synthesised via post-transcriptional modification processes.

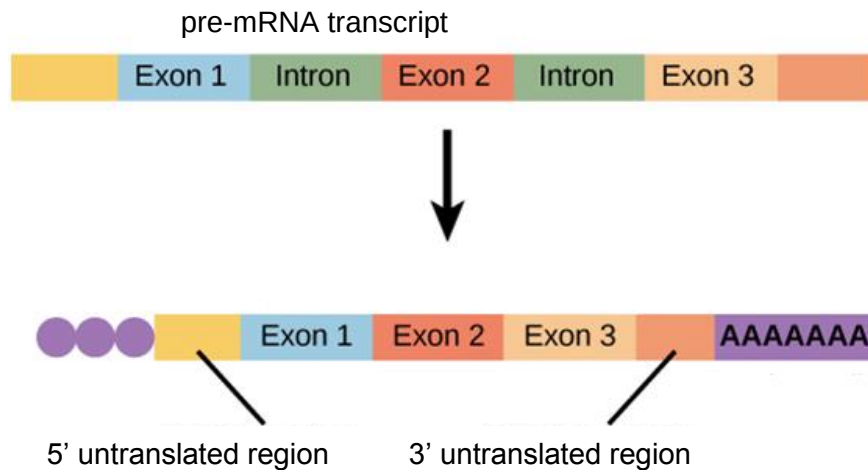


Fig. 4.2

(b) **Explain** why the post-transcriptional modifications are necessary.

RNA <u>splicing</u>	<u>Removal of introns sequences</u> and <u>splice of exons</u> together. R: removal of introns only; reject non coding regions are excised and coding regions are spliced, The specific terms are introns and exons. The word splice must be present. If RNA splicing is not specifically stated, the student must state that exons are spliced together. <i>Note that the picture did not show alternative splicing so any reference to more than one kind of polypeptide produced will not be awarded marks. But marks are given if the student stated that introns are excised.</i>
the <u>5' end</u> of the new RNA molecule is modified by addition of a <u>7-methyl guanosine cap</u> <u>R: 5' cap, 7-methyl guanosine cap without mention to the 5' end</u>	Any one of the below: 1. It helps to <u>protect the growing pre-mRNA chain from degradation by ribonucleases/exonuclease/enzymes.</u> 2. It facilitates the export of mature mRNA from the nucleus to the cytoplasm. 3. It is important for <u>recognition of the mature mRNA by ribosomes</u> to initiate translation.
<u>Polyadenylation:</u> The <u>addition of a long sequence of adenine nucleotides at the 3'end</u> of pre-mRNA to form a <u>poly(A) tail.</u> <u>R: poly A tail without mention to the 3' end</u>	<i>R: The poly(A) tail confers stability on mRNAs without any reason stated.</i> Any one of the below: 1. The poly(A) tail confers stability on many mRNAs, <u>increasing the time</u> during which the <u>mRNA remains intact and available for translation</u> before it is <u>degraded by cellular enzymes.</u> Or The poly(A) tail <u>protects mature mRNA</u> from being <u>degraded by RNases.</u> 2. It facilitates <u>the export of mRNA</u> from the <u>nucleus</u> to the cytoplasm. 3. It facilitates the attachment of ribosomes to the mature mRNA.

(c)

The mature insulin mRNA is translated to form a preproinsulin which is an inactive precursor. Describe one specific post-translational modification that converts it to its active form. [2]

Covalent modification such as **cleavage of a polypeptide chain** in insulin [1],
polypeptide C is cleaved and **chain A is connected to chain B** [1]

Or

Covalent modification such as **the formation of covalent bonds** in the polypeptide [1]

The formation of **two disulfide bonds** between **chain A and B** [1]

[Total: 8]

- 5 A transcription factor is a protein that promotes transcription at a particular locus. The gene *Ets 2* codes for a transcription factor, and is located on human chromosome 21. The gene is expressed in cartilage and bone tissues during development. In mice, *Ets 2* is located in chromosome 16 and mice with trisomy 16 show similar skeletal abnormalities to humans with Down's syndrome.

(a) (i) Explain the genetic basis of Down's syndrome in humans.

- 1 Presence of an extra copy of chromosome of 21/ $2n + 1$;
- 2 Due to translocation fragment 21 on to another chromosome/non-disjunction during meiosis;

(ii) Explain how a trisomy, such as trisomy 16 in mice, can occur in a zygote.

- 1 homologous chromosomes fail to separate at anaphase 1 /meiosis 1;
- 2 in oogenesis/formation of gametes in the mother where gamete has both homologous pair of chromosomes;
- 3 third chromosome added(from sperm) at fertilisation;

In order to investigate the possible role of *Ets 2* in Down's syndrome, transgenic mice have been produced which carry an extra copy of mouse *Ets 2*. These mice show similar skeletal abnormalities to mice with trisomy 16. Tissues from the transgenic mice produced up to 1.8 times more *Ets 2* messenger RNA than control mice.

(iii) Suggest why these transgenic mice and mice with trisomy 16 show similar abnormalities.

- 1 both have, three/an extra copy of *Ets 2*;
- 2 both produce more *Ets 2* messenger RNA than normal;
- 3 both have more transcription factor than normal

Max 2

Besides chromosomal mutations, gene mutations also occur in humans. Haemophilia A is characterized by deficiency in coagulation factor VIII, which results in prolonged oozing after injuries, tooth extractions, or surgery and renewed bleeding after initial bleeding has stopped. *F8* gene codes for the coagulation factor VIII.

In severe haemophilia A, spontaneous joint bleeding is the most frequent symptom. Serious complications can result from bleeding into the joints, muscles, brain, or other internal organs. Severity of symptoms can vary.

(b) Explain how substitution of bases in the *F8* gene could cause different levels of severity of symptoms in Haemophilia A.

- 1 Mutations involving substitutions of nucleotides in the *F8* gene lead to change in amino acid sequences, change in R groups, change in bonds formed and change in 3-D conformation and then the production of an abnormal/non-functional coagulation factor VIII
- 2 Depends where the mutation is. If altered protein effectiveness in the blood clotting process reduce but do not eliminate the activity of these proteins, which usually causes mild or moderate haemophilia.
- 3 Some mutations at essential sites → almost completely eliminate the activity of coagulation factor VIII i.e. the protein does not work at all → , resulting in severe haemophilia.

[Total: 10]

6 In summer squash, there are two pairs of alleles that determine fruit colour. The two genes are known to assort and segregate independently.

Two white-fruited plants are crossed. Both parents are known to be heterozygous for both genes. The cross produces the following offspring: 20 green-fruited plants, 58 yellow-fruited plants, and 218 white-fruited plants.

(a) State the name for this type of interaction between the two genes.

Epistasis (Accept: dominant epistasis; Reject: Recessive epistasis) [1]

- (b) Using the symbols D, d and E, e to draw a genetic diagram to explain the results given in the above information.

Parental phenotype	White-fruited plant	X	White-fruited plant
Parental genotype	DdEe	X	DdEe
Gametes	DE dE De de		DE dE De de

Gametes must be circled.

Punnet Square (with associated genotype)		DE	dE	De	de
	DE	DDEE White	DdEE White	DDEe White	DdEe White
	dE	DdEE White	ddEE Yellow	DdEe White	ddEe Yellow
	De	DDEe White	DdEe White	DDee White	Ddee White
	de	DdEe White	ddEe Yellow	Ddee White	ddee Green

Phenotypic ratio	White-fruited plant	Yellow-fruited plant	Green-fruited plant
	12	3	1

[4]

(c) Using enzyme production as a biochemical basis for fruit colour in squash, explain how the two genes interacted to produce the various phenotypes.

- i. Each gene codes for an enzyme that is involved in a **single biochemical pathway** that regulates the pigmentation in fruits of the squash plant.
- ii. The dominant allele, **D**, codes for an enzyme that will suppress / inhibit the action of the second enzyme thus result in no pigmentation of the fruits despite the nature of the gene (**E/e**).
- iii. The second gene (which has two alleles, **E** and **e**) codes for an enzyme that catalyse the production of either yellow pigments or green pigments that will be deposited in the fruits in the presence of recessive allele, **d**.

[OWTTE: Students are expected to write in their own words to show understanding of the concept of epistasis.]

A chi-squared test was carried out on the results of the cross.

Phenotype	Observed (O)	Expected (E)	$\frac{(O - E)^2}{E}$
green-fruited plants	20	19	0.1216
yellow-fruited plants	58	56	0.1126
white-fruited plants	218	222	0.0721
			$\chi^2 = 0.306$

Part of the critical values of the chi-squared distribution is shown below.

Degree of freedom	Probability								
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345

- (d) Explain how the chi-squared calculated value of 0.306 supports the hypothesis that the two genes assorted and segregated independently.
- i. At **p = 0.05** and at degree of freedom 2, the **χ^2 value is 0.306 is less than the critical value of 5.991** OR $\chi^2 = 0.306$ gives a **probability that is between 0.10 and 0.90 which is greater than p = 0.05.**
 - ii. There is **no significant difference** between the observed data and the expected numbers and any **deviations** of observed data from expected data are **due to chance**.
 - iii. The hypothesis that the two pairs of alleles assorted and segregated independently is **not rejected**.

[Total: 11]

- 7 It is important that the body's internal environment is controlled to maintain relative constancy of the internal environment despite external environmental changes. Maintaining a constant internal environment is called homeostasis. The nervous and endocrine systems both work to bring about this but their response patterns are different. Both the systems use chemical messengers to signal cells, but the speed at which these messages are transmitted is different.

- (a) Outline how the body receives information from the external environment to bring about a response. [3]

A **receptor detects changes** in the internal or external environment.

The **control centre receives information** from the receptors and initiates the response to maintain homeostasis.

An **effector is any organ or tissue** that receives information from the integrating center and acts to **bring about the changes** needed to maintain homeostasis.

Fig. 7.1 is a schematic diagram of the interaction between the different systems with the external environment.

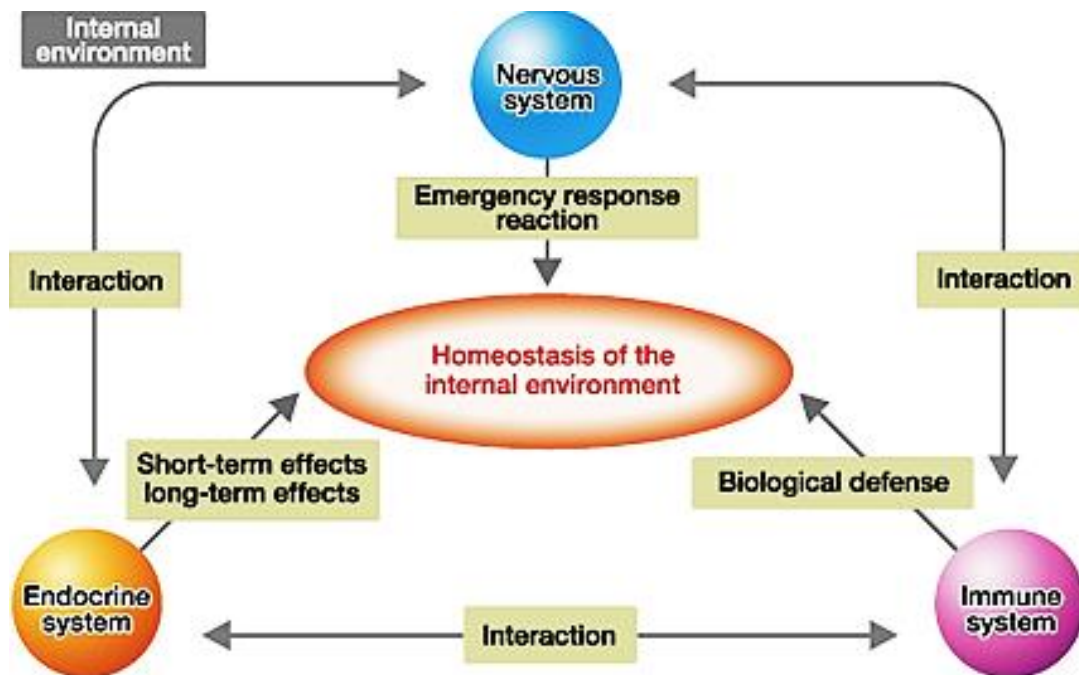


Fig. 7.1

- (b) With reference to Fig. 7.1, describe the important differences between endocrine and nervous systems.
- 1 The nervous system responds to stimuli **quickly / almost instantaneously** by **sending electrical action potentials along neurons**, which in turn transmit these action potentials to their target cells using **neurotransmitters/ electrical impulses**
 - 2 The endocrine system responds to stimuli **significantly slower**, as **hormones** must first be synthesized, **transported to their target cell**, and enter or signal the cell. / The target cell must go through the process of transcription, translation, and protein synthesis before the intended action of the hormone is seen.
 - 3 Although hormones act more slowly than a nervous impulse, their effects are **long lasting or short term**. The nervous system breaks down the neurotransmitter rapidly, the chemical messenger of the nervous system. This response to stimuli is **are short lived**.

Fig. 7.2 showed the steps needed to transmit a signal across a junction known as a synapse between two neurones.

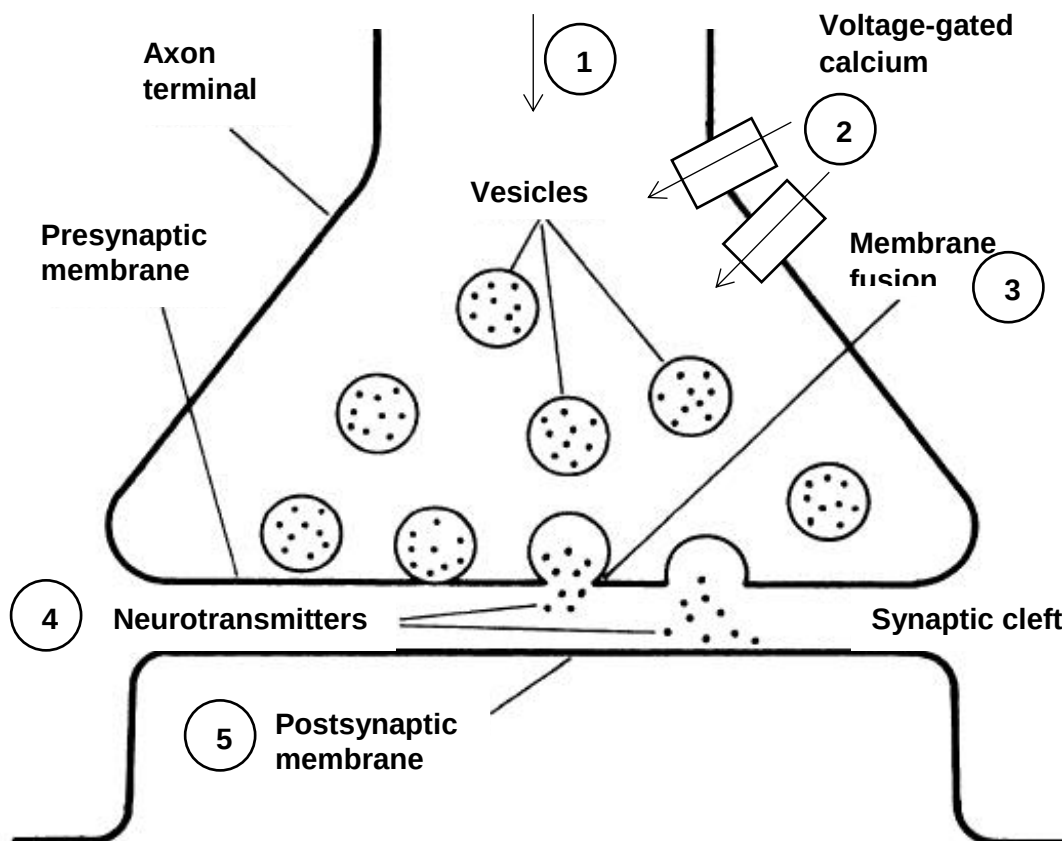


Fig. 7.2

- (c) (i) Explain which steps shown in Fig. 7.2 will result in synaptic delay.

Step 3: the release of the neurotransmitters bounded in vesicles via exocytosis/fusion with the presynaptic cell surface membrane requires time as movement to the CSM and the membrane fusion require time.
 Step 4: the diffusion of the acetylcholine through the synaptic cleft
 Step 5: the configuration change of receptor after the binding/the binding of the acetylcholine to the receptor requires time.

- (ii) Describe the roles of calcium ions in signal transmission across the synapse.

In this way, calcium influx during depolarization of the membrane of the presynaptic neurone regulates the presynaptic release of neurotransmitter substances.

This resulted in the movement of vesicles containing acetylcholine toward the presynaptic membrane. The vesicles then eject neurotransmitter into the synaptic cleft.

The synaptic knob releases a chemical transmitter, acetylcholine. Nerve gas prevents the breakdown of this chemical.

- (d) Suggest and explain one effect of nerve gas on the transmission of nerve impulse across the synapse.

The acetylcholine remains bounded to the receptor for extended period of time and the postsynaptic membrane will remain depolarised for an extended period, prolonging the effects.

Or

The acetylcholine cannot be broken down and recycled to the presynaptic cell will not be able to pass any chemical signals across the synapse due to this. There will be no action potential triggered in the postsynaptic cell /depolarisation of the postsynaptic membrane. [1]

- 8 Anoles are small, colour-changing lizards that are abundant in the Caribbean Islands. Hundreds of species live on the six islands of the Caribbean. Biologist Jonathan Losos has discovered the traits that enable dozens of anole species to adapt to different vertical niches in the forest of the islands.

While differences in limb length, body shape, and toepad size allow different species to flourish on the ground, on thin branches, or high in the canopy, changes in other characters, such as their colourful dewlaps, have played a key role in reproductive isolation and the formation of new species.

Fig. 8.1 shows a picture of *Anolis grahami*, commonly called Graham's anole. This is a species originally native to the island of Jamaica. Jamaica, part of the Caribbean islands, has a lush topography of mountains, rainforests and reef-lined beaches. Graham's anole is one of eight anole species found on Jamaica.



Fig. 8.1

- (a) Complete Table 8.1 to show the classification of *Anolis grahami*.

Kingdom	Animalia
Phylum	Chordata
Class	Reptilia
Order	Squamata
Family	Polychrotidae
Genus	<i>Anolis</i>
Species	<i>Anolis grahami</i>

[3]

Differences in the DNA sequences that encode cytochrome *b* in the eight species of Jamaican anoles have been measured.

(b) Suggest why the cytochrome *b* gene is used to measure changes in DNA sequences in closely related species.

- i. Cytochrome b gene is present in all the eight species and thus serves as a good basis for comparison in monitoring DNA changes across the eight different species;
- ii. Cytochrome b gene sequence changes very slowly OR most mutations in cytochrome b gene are silent as cytochrome b gene encodes for the cytochrome b complex that is essential for aerobic respiration. (Accept: highly conserved as it suggested that the change is slow / silent)
- iii. Changes in the cytochrome b DNA are also passed down the maternal line via mitochondrial DNA, thus no recombination occurs and any difference in DNA sequence between species is solely the result of mutations.

Fig. 8.2 shows differences in cytochrome *b* DNA plotted against time since colonisation (My).

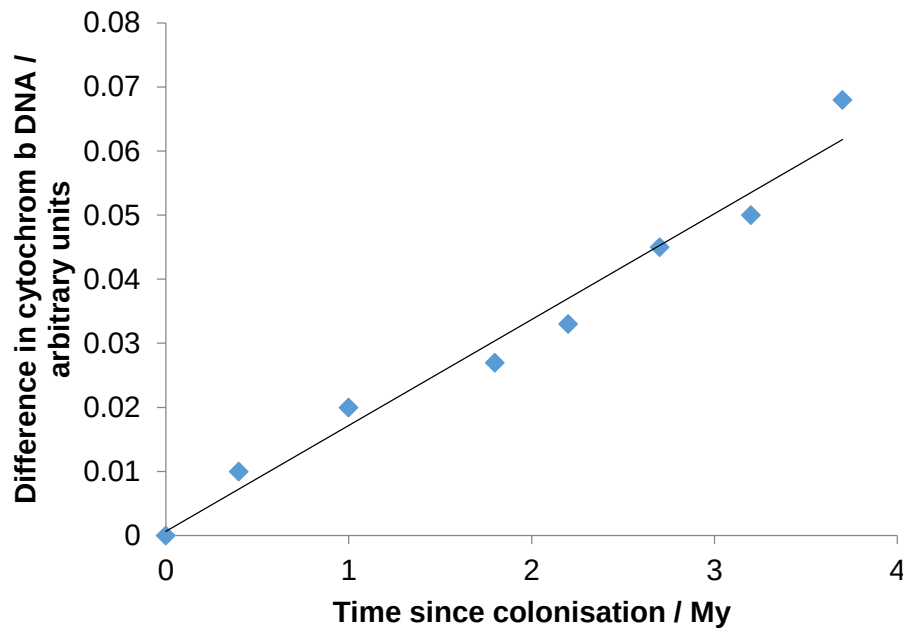


Fig. 8.2

- (c) Describe how these changes in DNA sequence of cytochrome *b* supports the neutral theory of evolution.
- i. Neutral theory of molecular evolution states that there is no selective advantage or disadvantage involved with changes on certain sets of DNA sequences, and there is no effect on the phenotype. (OWTTE such as does not affect Darwinian fitness / does not affect fitness)
 - ii. Mutations are silent; and there are a small number of changes that are not often observed in the phenotype. (Accept degeneracy of genetic code that allow mutations to be silent)
 - iii. The graph shown in Fig. 8.2 shows a straight line which reflects that the rate of mutation in cytochrome *b* gene sequences is steady / constant.

Using similarities in the banding of their chromosomes, it is possible to construct a phylogenetic tree for the evolution of these *Anolis* lizards.

Fig. 8.3 shows the phylogenetic tree for eight species of the *Anolis* lizards.

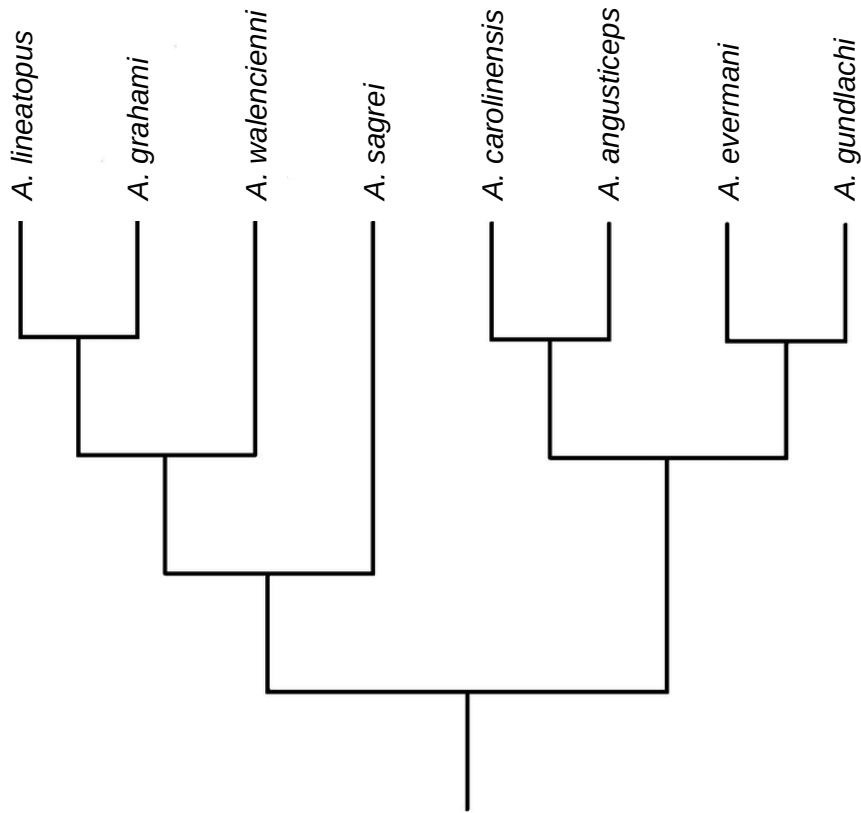


Fig. 8.3

- (d)** Explain the relationship between classification and phylogeny.

Classification is the organisation of species according to particular characteristics.

Classification may not take into consideration the evolutionary relationship between the species.

Phylogeny is the organisation of species according to particular characteristics which takes into consideration the evolutionary relationship between the species.

[Total : 11]

- 9 (a) Outline the three phases of the Calvin cycle [6]

Phase 1 (CO₂ fixation) /carboxylation

1. **Carbon dioxide acceptor is ribulose biphosphate (RuBP), a five-carbon sugar.**
2. **This is process is known as carboxylation / Carboxylation is catalysed by the enzyme RuBP carboxylase (RUBISCO).**
3. **The intermediate six-carbon product is unstable/This intermediate six-carbon product will be broken down to two molecules of glycerate 3-phosphate (GP) / 3-phosphoglycerate, a three-carbon organic acid.**

Phase 2 (Reduction)

4. **Glycerate-3-phosphate is phosphorylated by ATP to form 1, 3-bisphosphoglycerate.**
5. **1, 3-bisphosphoglycerate is reduced by NADPH to form glyceraldehyde 3-phosphate (G3P) / triose phosphate (TP).**

Phase 3 (Regeneration of CO₂ acceptor RuBP)

6. **Glyceraldehyde 3-phosphate (G3P) / triose phosphate has to be used to regenerate the RuBP.**
7. **ATPs are used to form glucose. Therefore, more ATPs are used in total compared to NADPH in Calvin cycle.**
8. **To form a glucose molecule (6-carbon), two G3P (3-carbon) are combined together.**

- (b) Explain the need for the production of genetically identical cells and fine control of replication. [7]
- 1 To ensure **genetic stability** as cell replicates via mitosis
New daughter cells contain the **full set of chromosomes** and **identical hereditary information** as those of the parent cell.
There is no variation in genetic information and the chromosome number
 - 2 **Asexual reproduction / vegetative propagation**
Mitotic division is the means for some plant and animal to produce asexually
 - 3 **Growth**
The number of cells within an organism increases by mitosis.
This is the basis of growth in multicellular organisms which originated from a single diploid zygote.
 - 4 **Cell replacement**
Cells in the skin, vaginal and oesophageal lining are constantly sloughed off, dying and being replaced by new ones.
When damaged tissues are repaired, the new cells must be exact copies of the cells being replaced so as to retain normal functions of cells.
 - 5 **Regeneration**
Some animals are able to regenerate whole parts of the body, such as legs in crustaceans and arms in starfish.

Max 3

1. Cell cycle can be out on hold at specific checkpoints. Allows cycle to be assayed for accuracy and can be halted if there are errors for repair or trigger apoptosis to occur.
2. Three main checkpoints (G1, G2 and M) act as control points where stop and go-ahead signals can **regulate the cell cycle**. /whereby there are sensitive mechanisms to detect internal state of the cell. Example nutritional state , DNA damage, cell size, DNA replication , chromosome attachment
3. A tumour suppressor gene called p53 gene produces a stop signal and plays key role in monitoring the integrity of DNA, producing proteins that halt cell cycle by regulating cyclin and cyclin dependent kinases interaction.
4. Mutation of the gene can cause dysregulation of the cyclin and cyclin dependent kinases interaction that interrupt the fine control of the checkpoints can lead to uncontrolled cell division.
5. This is because there is no detection of the presence of unreplicated or damaged DNA and hence lead to unrestrained proliferation of cells that could have DNA damage→ cancer cells. Hence, need for fine control of replication of DNA and cells for genetic stability and normal growth.
6. Ensured that replication occurs only when required in situations of growth , cell replacement so as to avoid wastage of resources and energy to replicate when not necessary .

And other AVP

Max 4

(c) Describe the role of NAD^+ in aerobic respiration.

[7]

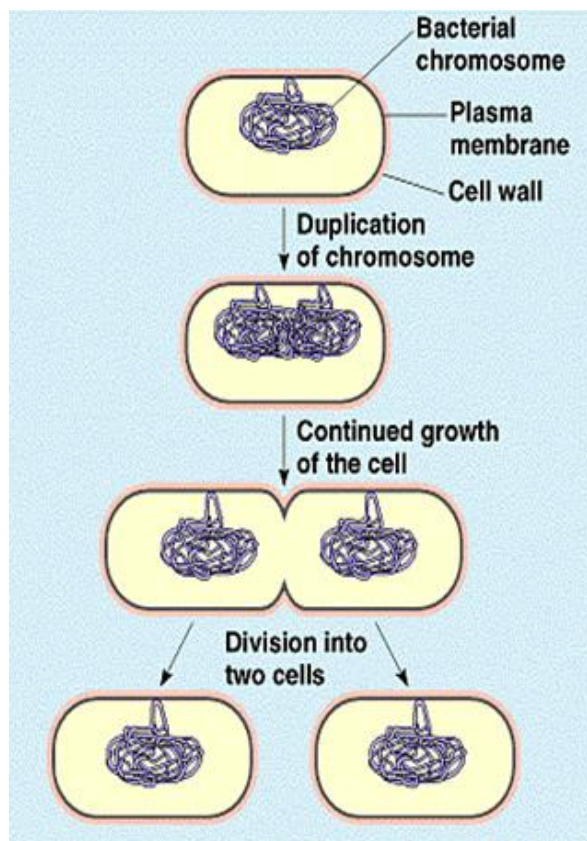
1. NAD^+ is a **coenzyme** involved in redox reactions and functions as
2. a **mobile electron (and proton) carrier** to carry high energy electrons and H^+ from organic molecules to the electron transport chain on the cristae of mitochondria;
3. **Organic molecules are oxidized** during glycolysis, link reaction and Krebs cycle and the **electrons and H^+** from the oxidation process are **transferred to NAD^+ to form NADH**;
4. The electrons in NADH are used to **reduce the electron acceptors** of the electron transport chain, while **NADH itself gets re-oxidised**;
5. As electrons pass down the chain, the release of energy in a series of redox reactions is coupled to the **phosphorylation of ADP to form ATP**;
6. The protons released in the oxidation of NADH is used to **establish the proton gradient** necessary for ATP synthesis; (accept chemiosmosis in place of proton gradient)
7. **1 NADH yields 2.5 ATP**
8. and **reoxidation of NADH** allows the **regeneration of the coenzyme NAD^+** , allowing it to pick up more protons and electrons from the Krebs cycle, link reaction and glycolysis so that these reactions can continue.

10 (a) Outline the process of binary fission in bacteria.

[6]

Ref to Genetics of Bacteria Lecture Notes

1. The bacterial chromosome begins its replication at a specific point called the **origin of replication**, made up of a particular sequence of nucleotide bases.
2. The double helix unwinds to form two separate strands and the overall direction of replication is outward from the origin, in both directions (i.e. **bidirectional**) resulting in an **interlocking structure** made up of 2 daughter molecules is formed.
3. **Topoisomerase** is needed to cut, separate and reseal the two DNA molecules and the duplicated DNA molecules are attached to the plasma membrane.
4. The cell **elongates** in preparation for cell division (to approximately two times its size).
5. When the cell divides, it separates the duplicated DNA and the cell membrane folds inwards (**invaginates**) to form a double layer across the long axis of the cell. (Reject: cell wall invaginate, cell invaginate)
6. **New cell wall** layers are formed within the membrane layers and this divides the cell into two smaller identical cells that may stay together or separated (dividing cells).



- (b)** Explain the effects of competitive and non-competitive inhibitors on the rate of enzyme activity. [6]

Ref to Enzymes tutorial answers

Competitive inhibitor

1. It has close structural resemblance to the substrate, thus it competes with substrate molecule for active site of enzyme.
2. It binds to active site of the enzyme hence the rate of reaction/enzymatic activity is reduced as inhibitor excludes substrate molecules from binding.
3. The effect of competitive inhibitor can be reduced by increasing the substrate concentration which increases the probability of enzyme-substrate collision rather than enzyme-inhibitor collision.
4. Therefore, the rate of reaction increased and the effect of inhibitor is negligible. Rate of reaction will reach maximum velocity.

Max **3-**

Non-competitive inhibitor

1. Non-competitive inhibitor has no close structural resemblance to the substrate thus it binds to a site other than the active site.
2. Attachment of inhibitor either inhibits attachment of substrate to active site by changing the shape of the active site or prevents catalysis after substrate is bound to the enzyme.
3. Inhibition cannot be overcome by high substrate concentration; therefore probability of enzyme-substrate collision remains low.
4. Therefore, the rate of reaction decreased and will not reach maximum velocity; the effect of inhibitor cannot be reduced.

Max **3**

(c) Describe the causes of genetic variation in a population. [8]

1. Independent assortment and segregation of homologous chromosomes occurs during metaphase I and anaphase I, respectively.
2. During metaphase I, the homologous chromosomes, each consisting of one maternal and one paternal chromosome are **randomly arranged at the equator** OR the arrangement of one pair of homologous chromosomes at the equator is **independent of the other pairs of homologous chromosomes**.
3. During anaphase I, the chromosomes of one homologous pair will **separate independently** of the other pairs that result **in different combinations of chromosomes** in the daughter cells at the end of meiosis I.
4. Crossing over of segments of non-sister chromatids in prophase (I) of homologous chromosomes at prophase I will lead to **new combinations of alleles** (REJECT: genes) on chromosomes of the gametes.
5. The 2 replicated homologous chromosomes form **synapsis / pair up** during prophase I of meiosis whereby they are joined at several points along their length such that **non-sister chromatids overlap and intersect**.
6. Breakage of segments of non-sister chromatids and **subsequent recombination of segments of non-sister chromatids** of homologous chromosomes at each chiasma allows mutual exchange of corresponding segments of alleles.
7. During sexual reproduction, random fertilisation of gametes occurs whereby a random sperm will fertilise the ovum resulting in further variation within a population.
8. **Mutations in the genes** in gametes immediately results in the **formation of new alleles** thus changes the gene pool of a population by substituting one allele for another.